

The Goals of Psychologists

What do the various kinds of psychologists do?

What are their guiding assumptions about the causes of behavior; the relation of brain activity to experience, and the roles of nature (heredity) and nurture (environment)?

Years ago, one of my students asked me, “When will we get to the kind of psychology we can ‘use’ on people?” Another student asked me whether I, as a psychologist, had any tips to offer him on how to seduce his girlfriend. (I don’t. If I did, I wouldn’t tell him, for ethical reasons. If I did know powerful tricks to control people’s behavior and had no ethics, I would probably have a different job than college professor!) Psychologists do not seek techniques for manipulating or tricking people. They *do* try to understand why people act the way they do, to help people to better understand themselves, and also to help people change their behavior to achieve their own goals.

Broadly defined, **psychology** is *the systematic study of behavior and experience*. The term *psychology* derives from the Greek roots *psyche*, meaning “soul” or “mind,” and *logos*, meaning “word”; thus, psychology is literally the study of the mind or soul. Every culture has developed a “folk psychology” to try to explain people’s feelings and actions. As an example, the *Illongot* people of the Philippines believe in a concept called *rinawa* (roughly, a “life force”) that is responsible for a person’s energy and health. They believe that *rinawa* gradually leaves the body throughout life, so even by age 20 one has much less *rinawa* than a child has. During dreams, *rinawa* leaves the body and travels around. If the *rinawa* happens to eat with a dead person during a dream, then it will start to feel comfortable among the dead, and the person will in fact begin to die (Lillard, 1997).

These assumptions seem very natural to the *Illongot*, less so to most other people. But we also make assumptions that we take for granted. Some suggest that psychology is nothing more than common sense. Well, if so, psychologists need not be ashamed. Most people spend a great deal of time trying to understand their own and other people’s behavior, and common sense is much more accurate about

psychology than it is about, say, physics. But your common sense is probably not the same as that of the *Illongot*. Every culture, and I suppose even every individual, has different notions of common sense, and we all need to evaluate those ideas. When we do, we find that some of our common sense ideas are correct and some are not.

In this text, I shall be much more concerned with evaluating the ideas that are common among the probable readers of this book than exposing the possible weaknesses of, for example, *Illongot* folk psychology. Here we shall be concerned primarily with psychological issues as they relate to life in a technological society and as they relate to the philosophical concerns that have traditionally interested people in Western cultures. Psychologists in other societies, such as the rural parts of India, for example, have different concerns (Gergen, Gulerce, Lock, & Misra, 1996). It is completely natural for each society to focus on different issues, although at times it is helpful to compare notes. Sometimes people from a different culture raise questions and challenge ideas that we might otherwise take for granted.

Major Philosophical Issues in Psychology

Many psychological concerns date back to the philosophers of ancient Greece. Although psychology has moved away from philosophy in its methods, it continues to be motivated by some of the same issues. Three of the most profound questions are free will versus determinism, the mind-brain problem, and the nature-nurture issue.

Free Will Versus Determinism

Beginning with the Renaissance period in Europe, people began to look for scientific explanations for the phenomena they observed. One of the key points of this scientific revolution was a shift toward seeking the *immediate* causes of an event (what led to what) instead of the *final* causes (the ultimate purpose of the event in an overall plan). Scientists analyzed the motion of objects in terms of pushes and pulls and other laws of nature (White, 1990). That is, they made an assumption called **determinism**, *the assumption that everything that happens has a cause, or determinant, in the observable world*.

Is the same true for human behavior? Each of us is, after all, part of the physical world. Your brain is made of chemical compounds that are subject to the same laws of nature as anything else. According to the *determinist* assumption of human behavior, everything we do has a cause (Figure 1.1).

Clearly, at least some of these causes lie within us. A person walking down a mountainside is not the same as a rock that rolls or bounces down the same mountainside. The claim of psychological determinism is that, even when



FIGURE 1.1 Behavior is guided by external forces, such as waves, and by forces within the individual. According to the determinist view, even those internal forces follow physical laws.

you make complicated decisions about how to get down a mountain safely, your decision is a product of the combined influence of your genetics, your past experiences, and the current environment (Sappington, 1990). Just as an engineer can design a robot to consider information and make appropriate decisions, your genes and your experience have programmed you to make appropriate decisions. (You did not design or program yourself.)

Logically, the opposite of determinism should be *indeterminism*, the idea that events happen randomly with no cause at all. In psychology, however, few people argue that behavior is random or indeterminate. Opponents of determinism instead defend **free will**, generally defined as the *belief that behavior is caused by a person's independent decisions, not by external determinants*. But what are “independent decisions”? If we are talking about an “internal” decision made by the individual, then there may be no difference between the determinist and free will positions; determinism agrees that the individual makes decisions but points out that a robot does too. To be really different from determinism, the free will position must hold that a person’s “independent decision” is independent from the physics and chemistry of the body.

The test of determinism is ultimately empirical: If everything we do has a cause, our behavior should be predictable. To some extent, it is. For example, suppose that students studying in various rooms of a campus building hear an announcement: “A fire has broken out in this building. By the time we bring the fire under control, the smoke and fumes may become hazardous. We therefore request that everyone calmly leave the building.” We can predict that almost everyone will promptly leave the building. We can make an even more accurate prediction if we know something about the individual students: Everyone will leave the building *except* those who could not hear the announcement, those who do not understand English, and those who have been advised by a friend “every year at about this time someone says to leave the building because

of a fire, but it’s just a silly exercise to show that psychologists can predict your behavior, so ignore the warning.”

In other situations, however, you might object that no one could possibly predict your behavior, no matter how much they knew about you. For example, no one could accurately predict what you will choose to eat for lunch tomorrow or which color sweater you will buy or how many pages of this book you will read before you quit to do something else.

You are right; psychologists will probably never be able to predict every detail of your behavior; however, this unpredictability does not imply an absence of causes. Physicists and mathematicians today talk about *chaos*, the complex effects that result from many small influences. For example, imagine that I drop a golf ball at the top of a hilly road and measure the exact point at which it eventually comes to rest. Then I take the ball back to the same location at the top of the hill and drop it again the same way—or as close to the same way as I can. Will the ball eventually land in the same place as it did the first time? No, the first bounce or two will land in nearly the same place as before, but with each succeeding bounce, the ball will veer further and further from its original route, reflecting the cumulative effect of an enormous number of tiny influences. Similarly, when you are deciding what to eat for lunch or which sweater to buy, your behavior is subject to so many tiny influences that your choice may be no more predictable than the final bounce of that golf ball. This unpredictability stems from a great number of small causes, not from a lack of causes.

Let me concede an important point, however. In the introduction to this chapter I said that we all sometimes state something with confidence that is really only an assumption. This is such a case. Researchers assume that every behavior has a natural cause, because that assumption seems to work and because the only way to test the assumption is to see how far we can go with it before we find some limits. Still, to be honest, it is merely an assumption, not a certainty.

SOMETHING TO THINK ABOUT

What kind of evidence, if any, would support the concept of free will? Demonstrating that a particular theory makes incorrect predictions about behavior under given circumstances would not refute determinism. To support the concept of free will, one would need to demonstrate that no conceivable theory could make correct predictions. Should a psychologist who believes in free will conduct the same kind of research that determinists conduct, or a different kind of research, or no research at all? ★

The Mind-Brain Problem

Every movement we make depends on muscle activity controlled by the nervous system, and every sensory experience depends on the activity of the nervous system. All activities of the nervous system follow the laws of physics and

chemistry. What, then, is the role of the mind? *The philosophical question of how experience is related to the brain* is the **mind-brain problem** (or mind-body problem). Does the brain produce the mind? If so, how and why? Or does the mind control the brain? If so, how can a nonphysical entity control a physical substance? Or are the mind and the brain just two names for the same thing? If so, what does it mean to say that they are the same?

Although the mind-brain problem is a particularly difficult philosophical issue, it does lend itself to research. Research can determine links between brain activity, on the one hand, and behavior and experience, on the other hand. For example, consider Figure 1.2. A technique called positron-emission tomography enables investigators to measure the amount of activity in different parts of the brain at various times. The photos in Figure 1.2 show brain activity while a person is engaged in nine different tasks. Red indicates the highest degree of brain activity, followed by yellow, green, and blue. As you can see, the various tasks tend to activate different areas of the brain, although all areas show at least some activity at all times (Phelps & Mazziotta, 1985).

Data such as these show a very close relationship between brain activity and psychological events. You might well

ask: Does this mean that brain activity caused the associated thoughts, or did those thoughts cause that pattern of brain activity? Many brain researchers would reply that neither brain activity nor mental activity “causes” the other; rather, brain activity and mental activity are the same. (See Dennett, 1991.)

Even if we accept this position, we are still far from understanding the mind-brain relationship fully. Is mental activity associated with all brain activity or just certain types? Why is there such a thing as conscious experience at all? Could a brain get along without it?

Research studies are not about to resolve the mind-brain problem and put philosophers out of business. But research results do constrain the types of philosophical answers that we can seriously consider. The hope of learning more about the mind-brain relationship is one of the ultimate goals for many psychologists, especially those whose work we shall study in Chapters 3 and 4.

SOMETHING TO THINK ABOUT

One way to think about the mind-brain relationship is to ask whether something other than a brain—a computer, for example—could have a mind. How would we know?

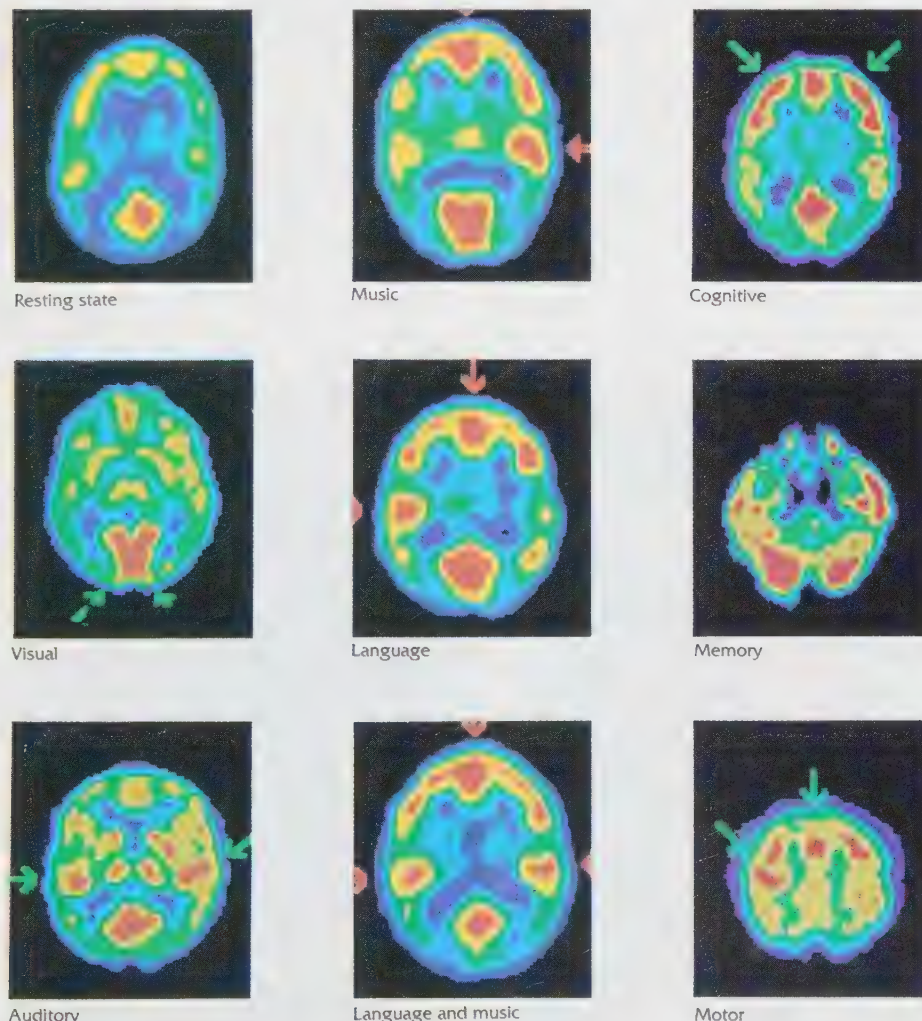


FIGURE 1.2 PET scans show the brain activity of normal people engaged in different activities. Left column: Brain activity with no special stimulation, while passively watching something or listening to something. Center column: Brain activity while listening to music, language, or both. Right column: Brain activity during performance of a cognitive task, an auditory memory task, and the task of moving the fingers of the right hand. Red indicates the highest activity, followed by yellow, green, and blue. Arrows indicate the most active areas. (Courtesy of Michael E. Phelps and John C. Mazziotta, University of California, Los Angeles, School of Medicine.)

What if we built a computer that could perform all the intellectual functions that humans perform? Could we then decide that the computer is conscious, as human beings are? ★

The Nature-Nurture Issue

Why do most little boys spend more time than little girls do with toy guns and trucks and less time playing with dolls? Are such behavioral differences mostly the result of genetic differences between boys and girls, or are they mostly the result of differences in how society treats boys and girls?

In many countries, alcoholism is a serious problem. In other countries—Turkey, for example—alcohol abuse is rare. Are these differences entirely a matter of social custom, or do certain genes influence how much alcohol people consume?

Certain psychological disorders are more common in large cities than in small towns and in the countryside. Does life in crowded cities somehow cause psychological disorders? Or do people develop such disorders because of a genetic predisposition and then move to a big city because that is the best place to find jobs, housing, and welfare services?

Each of these questions is related to the **nature-nurture issue** (Figure 1.3): *What are the roles of heredity and environment in the development of various behaviors?* Obviously, people need both heredity and environment to develop at all. However, the *differences* between one person and another may depend mostly on differences in heredity or differences in environment. The nature-nurture issue shows up from time to time in practically all fields of psychology, and it seldom has a simple answer.

SOMETHING TO THINK ABOUT

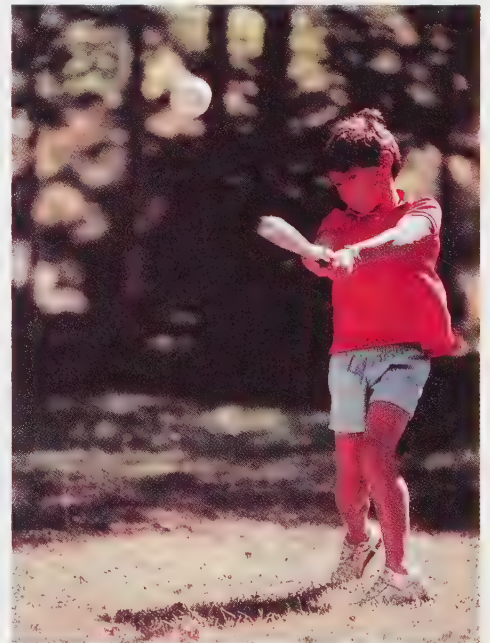
Suppose researchers found that the difference in the rate of alcohol abuse between Turkey and most other countries is due to environmental causes (the strict legal sanctions against alcohol in Turkey). Should we then assume that the differences in alcohol use among other countries is also due to environmental causes? ★

What Psychologists Do

We have presented some major philosophical issues related to the entire field of psychology. However, few psychologists are ambitious enough to attempt to answer these ultimate questions; they usually deal with more specific or more detailed issues.

Psychology is an academic, nonmedical discipline that includes many branches and specialties, ranging from the helping professions to research on learning, memory, brain functions, and so forth. The educational requirements for becoming a psychologist vary from one country to another (Newstead & Makinen, 1997). In the United States and Canada, a psychology student gets a bachelor's degree (usually after four years of college) and then probably a Ph.D. degree (at least another four or five years, sometimes much more). Some people practice psychology with a master's degree (intermediate between a bachelor's and a doctorate), and others have a Psy.D. (Doctor of Psychology) degree, which generally requires less research experience than a Ph.D. Any psychologist specializes in a particular branch of psychology, such as experimental, developmental, clinical, or industrial.

FIGURE 1.3 Why do different children develop different interests? They may have had different hereditary tendencies, but they have also experienced different environmental influences. Separating the roles of nature and nurture can be difficult.



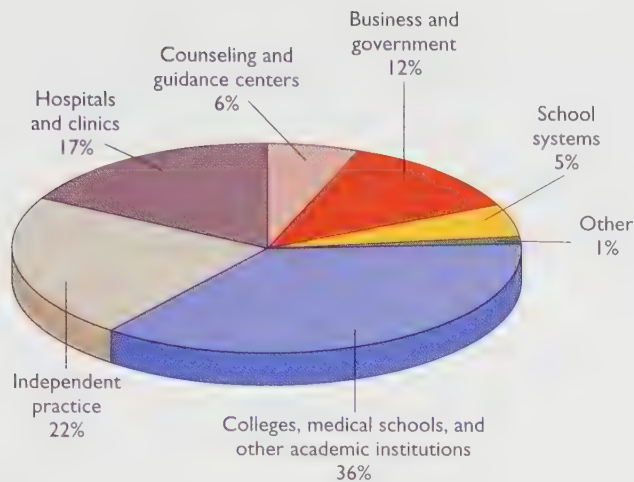


FIGURE 1.4 More than one third of psychologists work in academic institutions; the remainder find positions in a variety of settings.

Psychologists work in many occupational settings, as shown in Figure 1.4. A little more than a third work in academic institutions—colleges, universities, and medical schools. Almost 40% work in health-provider settings—-independent practices, hospitals, and clinics. Others work in business, government, guidance and counseling centers, and public school systems.

Psychologists in Teaching and Research

Many psychologists who are not clinical psychologists have positions in colleges and universities where they teach and do research that will, they hope, lead to a greater understanding of behavior and experience and perhaps have useful applications. A small percentage of psychologists work in full-time research institutions. Here, we preview a few major categories of psychological research.

Biological Psychology A **biopsychologist** (or **behavioral neuroscientist**) *tries to explain behavior in terms of biological factors, such as electrical and chemical activities in the nervous system, the effects of drugs and hormones, genetics, and evolutionary pressures.* For example, the evidence suggests that certain genes influence the probability that someone will develop schizophrenia, depression, or alcoholism. These genes do not completely control the outcome; the final results depend on experience as well as genes.

Biopsychologists also study the effects of brain damage. Brain damage can result from such things as a sharp blow to the head, a ruptured blood vessel in the brain, an interruption of oxygen supply, prolonged malnutrition, or exposure to toxic chemicals. The effects of brain damage on behavior vary enormously, depending on the location and extent of the damage.

People have long known that various drugs can alter behavior. For example, opiates (such as heroin or morphine)

generally make people quiet, passive, and insensitive to pain. Amphetamines and cocaine stimulate increased activity in most people. Biopsychologists try to understand how drugs affect the brain. They have found that most drugs that affect behavior do so by altering the chemical communication between one neuron and another at junctions called *synapses*.

So, according to biopsychologists, why are people different from each other? There are many reasons: People are born with different genes; they develop slightly different brains and different hormonal patterns. Some have suffered brain damage; some are under the influence of drugs or of nutritional deficiencies that affect the brain. Anything that affects the body, especially the brain, will also affect behavior.

Learning and motivation The research field of **learning and motivation** studies *how behavior depends on the outcomes of past behaviors and on current motivations.*

How often we engage in any particular behavior depends on the results of engaging in that behavior in the past. For example, you undoubtedly have a friend whose interests differ from yours. One of you likes dancing; the other likes painting. Or one likes to play the guitar, and the other is active in political activities. The one who likes dancing may have received praise for dancing, or may experience some good bodily sensations while dancing. The one who enjoys painting may once have tripped while dancing. The details vary from one person to another, but our current behavior depends on our past learning.

Behaviorists do research to determine how the consequences of an action modify future behavior. For example, do frequent payoffs (money, praise, etc.) produce the same results as less frequent or less predictable rewards? If someone were expecting a large payoff and instead got a smaller payoff, what would be the effect, compared to someone who was used to a smaller payoff? What are the effects of punishment? Notice the theoretical orientation: The behaviorist studies what the person *does* as a result of the consequences of past actions, not what the person *thinks*. Because of this emphasis on actions instead of thoughts, behaviorists can conduct many of their studies on nonhuman animals.

Cognitive Psychology **Cognition** refers to *thinking and acquiring knowledge.* A **cognitive psychologist** *studies those processes.* (The root *cogn-* also shows up in the word *recognize*, which literally means “to know again.”) As a rule cognitive psychologists do not simply ask people to describe their thought processes. (If people understood their own thoughts that well, there would be less need for psychologists.) Cognitive psychologists conduct experiments to infer what people know, how they came to know it, and how they use their knowledge to solve new problems. Some cognitive psychologists develop and test computer models of how people think.

One typical question for a cognitive psychologist: What do experts know or do that sets them apart from

other people? One possible distinction is simply that the expert knows more facts. Consider a subject on which you are an expert: how to find your way around your college campus. A fellow student asks you, “How do I get from here to the biology building?” To answer, you can draw on the knowledge you share with the other student: “Go over toward the library. Then cut behind the library, between the library and the math building. The biology building will be right in front of you.” Now, suppose that a visitor who has never been on campus before asks you the same question. You say, “Go over toward the library . . .” “Wait, where’s the library?” “Well, go out this door, make a right, go to the next street . . .” Someone with little or no previous knowledge will need detailed and extensive instructions (Isaacs & Clark, 1987).

Another distinction between the expert and the nonexpert is that the expert can identify more categories. For example, someone might look at a group of birds on a beach and say, “Hey, look at all the seagulls.” An expert bird-watcher might reply, “There are three gull species and two tern species.” Moreover, the expert can identify the *right* categories. The inexperienced bird-watcher might say, sheepishly, “Oh, I see. Some of them have darker feathers than others.” The expert would reply, “No, the ones with darker feathers are just younger. To tell one species from the other you have to check the color of the beak, the color of the legs, the size of the bird, the color of the eyes . . .” The expert knows what to look for—what is relevant and what is not (Murphy & Medin, 1985).

Developmental Psychology Developmental psychologists study the behavioral capacities typical of different ages and how behavior changes with age, “from womb to tomb.” In a typical study, developmental psychologists examine a particular behavior across a certain age span, such as language from age 2 to age 4, or the speed of solving intellectual tasks from age 60 to age 80. The first



Developmental psychologists study the behavioral differences among people of different ages.

question is: What do people do at one age that they do *not* do at another age? The second question is: Why? Was the change due to a biological process, or to changes in experience, or to a complex combination of both? Developmental psychologists frequently need to address the nature-nurture issue that we mentioned earlier.

Social Psychology Social psychologists study how an individual influences other people and is influenced by others. When we are with other people, we tend to take our cues from them about what we should do. Suppose you arrive at a party and notice that the other guests are walking around, talking, and helping themselves to snacks. You do the same. When you go to a religious service or an art museum, you notice how other people are acting and, again, conform your behavior to theirs.

Certainly, if you had grown up in a different country, you would have developed vastly different customs. Even within a given culture, though, people acquire different behaviors because of the people around them. If you had made friends with a different set of people in high school, you might be a much different person today. According to social psychologists, people are also heavily influenced by other people’s expectations. For example, parents often intentionally or unintentionally convey expectations that boys will be more competitive and girls will be more cooperative, or that teenagers will be immature and that 25-year-olds will be responsible. At least to some extent, people’s behavior tends to live up to—or down to—the expectations of others. Social psychologists study such influences.

CONCEPT CHECK

1. (a) Of the kinds of psychological research just described—biological psychology, learning and motivation, cognitive psychology, developmental psychology, and social psychology—which field concentrates most on children? (b) Which is most concerned with how people behave in groups? (c) Which concentrates most on thought and knowledge? (d) Which would be most interested in the effects of brain damage? (e) Which is most concerned with studying the effect of a reward on future behavior? (Check your answers on page 16).

Clinical Psychologists and Other Psychotherapists

When most people hear the term *psychologist*, they first think of *clinical psychologists*, who constitute one type of **psychotherapist**, specialists in helping people with psychological problems ranging from depression or other serious disorders to marriage conflicts, difficulties making decisions, or even the mere feeling that “I should be getting more out of life.” Some clinical psychologists are college professors and researchers, but most are full-time private practitioners.

It is important to distinguish between clinical psychologists and psychiatrists. **Clinical psychologists** have an *advanced degree in psychology, with a specialty in understanding and helping people with psychological problems*. As part of their training, they undergo at least one year of supervised clinical work called an *internship*. Clinical psychologists can base their work on any of various theoretical viewpoints, which we shall explore in later chapters. They try, in one way or another, to understand why a person is having difficulties and then to help that person to become more successful and to make better choices.

Psychiatry is a *branch of medicine that deals with emotional disturbances*. To become a psychiatrist, a student first earns an M.D. degree and then takes an additional four years of residency training in psychiatry. Psychiatrists and clinical psychologists provide similar services for most clients: They listen, ask questions, and try to help. Psychiatrists, however, are medical doctors and can therefore prescribe drugs, such as tranquilizers and antidepressants, whereas psychologists cannot.

Does psychiatrists' ability to prescribe drugs give them an advantage over psychologists? Sometimes, but not always. Ours is an overmedicated society. Some psychiatrists habitually treat anxiety and depression with drugs, whereas a psychologist tries to treat problems by changing the person's way of living. Some psychologists favor a change in the law that would permit them to take some extra training to gain the right to prescribe drugs. Others fear that gaining prescription privileges would mean losing psychologists' distinctive focus on talking extensively with their clients (Hayes & Helby, 1996). Prescription privileges will probably remain a controversial topic.

Several other kinds of professionals also provide help and counsel. Psychiatric nurses and psychiatric social work-

ers have an undergraduate or master's degree in nursing or social work plus additional training in the care of emotionally troubled people. **Psychoanalysts** are *psychotherapists who rely heavily on the theories and methods of the early 20th-century Viennese physician Sigmund Freud*. There is some question about who may rightly call themselves psychoanalysts. Some people apply the term to any psychotherapist who relies heavily on Freud's methods; others apply the term only to graduates of a six- to eight-year program at an institute of psychoanalysis. Those institutes admit only people who are already either psychiatrists or clinical psychologists. Thus, people completing psychoanalytic training will be at least in their late 30s.

Table 1.1 compares various types of psychotherapists.

CONCEPT CHECK

- Can psychoanalysts prescribe drugs? (Check your answer on page 16.)

Nonclinical Fields of Applied Psychology

Psychologists also work in business, industry, and school systems in some capacities that might be unfamiliar to you, doing things you might not think of as psychology. The job prospects in these fields have been quite good, however, and you might find these fields interesting. Let's explore these fields in detail here, because they do not have whole chapters devoted to them later, as do areas such as developmental and social psychology.

Industrial/Organizational Psychology *Industrial/organizational psychology is the study of people at work*. It deals with such issues as how to match the

TABLE 1.1 Clinical Psychologists and Other Psychotherapists

TYPE OF THERAPIST	EDUCATION	NUMBER ACTIVELY PRACTICING IN U.S.*
<i>Clinical psychologist</i>	Ph.D. with clinical emphasis, or Psy.D., plus internship. Total of generally 5+ years after undergraduate degree.	42,000
<i>Psychiatrist</i>	M.D. plus psychiatric residency. Total of 8 years after undergraduate degree.	33,000
<i>Psychoanalyst</i>	Psychiatry or clinical psychology plus 6–8 years in a psychoanalytic institute. Others who rely on Freud's methods also call themselves psychoanalysts.	9,000
<i>Psychiatric nurse</i>	From 2-year (A.A.) degree to master's degree, plus supervised experience.	16,000
<i>Clinical social worker</i>	Master's degree plus 2 years of supervised experience. Total of at least 4 years after undergraduate degree.	45,000

*Based on data from Howard et al. (1996) and estimates provided by the American Psychoanalytic Association, the American Psychiatric Nurses Association, and the National Association of Social Workers.

right person with the right job, how to train people for jobs, how best to determine salaries, and how to organize the workplace so that workers will be both productive and satisfied. Industrial/organizational psychologists call upon their knowledge of social psychology, cognitive psychology, and standardized testing.

Here's one example of a typical concern for industrial/organizational (I-O) psychologists (Campion & Thayer, 1989): A company that manufactures complex electronic equipment needed to publish reference and repair manuals for its products. The engineers who designed the devices could not devote the necessary time to write the manuals, and none of them were skilled writers anyway. So the company hired a technical writer to prepare the manuals, but after a year she received an unsatisfactory performance rating. The manuals she wrote contained too many technical errors; besides that, she was constantly complaining.

The writer countered that, when she asked various engineers in the company to check her manuals or to explain technical details to her, they were always too busy. She found her job complicated and frustrating; her office was badly lit, noisy, and overheated, and her chair was uncomfortable. Whenever she mentioned any of these problems, however, she was told that she "complained too much."

In a situation like this, an industrial/organizational psychologist can help the company to evaluate the possible solutions. First, the problem might be employee selection; maybe the company hired the wrong person for this job. If so, they should fire the current writer and hire someone who is an expert on electrical engineering, who is also an outstanding writer, and who *likes* a badly lit, noisy, overheated, uncomfortable office. However, if the company cannot find or afford such a person, then the company needs to improve the working conditions and provide the current employee with more training or more help with the technical aspects of the job.

Here is another example of what an industrial/organizational psychologist does: A plywood plant required workers to align strips of wood on a moving belt (Campion & Thayer, 1985). They had to bend on one leg and reach out to move the strips while balancing on one leg and operating a foot pedal with the other. The supervisor was dissatisfied with their performance and grumbled that "workers today are lazy and inefficient." In fact, given how the task was designed, no one could have performed it comfortably. After I-O psychologists helped the company to redesign the equipment, workers were able to perform the task more effectively and to do so with fewer complaints.

As these two examples show, when a company blames its workers for doing a poor job, I-O psychologists often discover that the real problem lies with the job itself. In some cases, however, the problem does lie with the workers. In such cases, the I-O psychologist would try to improve the company's method of selecting employees, such as identify-

ing useful cognitive tests (similar to the tests students take in school), personality tests, improved interviews, biographical inventories, and so forth.

SOMETHING TO THINK ABOUT

I-O psychologists usually consult with business and industry, but suppose they were called upon to help a university where certain professors had complained that "the students are too lazy and stupid to understand the lectures." How might the I-O psychologists react? ★

Ergonomics Several years ago, my son Sam, who was then about 16 years old, turned to me as he was rushing out the door and asked me to turn off his stereo. I went to the stereo in his room and tried to find an "on-off" switch or a power switch. No such luck. I looked in vain for the manual. Finally, in desperation, I simply unplugged the stereo.

Learning to operate our increasingly complex machinery is one of the perennial struggles of modern life. Sometimes, the consequences can be serious. Imagine an airplane



Ergonomists help redesign machines to make them easier and safer to use. An ergonomist uses principles of both engineering and psychology.

pilot who intends to lower the landing wheels and instead raises the wing flaps. Or a worker in a nuclear power plant who fails to notice a warning signal.

In one specialized field of psychology, an **ergonomist**, or **human factors specialist**, *attempts to facilitate the operation of machinery so that the average user can use it as efficiently and as safely as possible*. The term *ergonomics* is derived from Greek roots meaning “laws of work.” Ergonomics was first used in military settings, where complex technologies sometimes required soldiers to spot nearly invisible targets, to understand speech through deafening noise, to track objects in three dimensions while using two hands, and to make life-or-death decisions in a split second (Taylor, 1957). The military turned to psychologists to determine what skills their personnel could master and to redesign the tasks to fit those skills.

Ergonomists soon applied their experience not only to business and industry, but also to everyday devices. As Donald Norman (1988) pointed out, many intelligent and educated people find themselves unable to use all the features on a camera, a microwave oven, or a videocassette recorder; some even have trouble setting the time on a digital watch.

When designing machinery, ergonomists emphasize the **principle of compatibility**, *the concept that people's built-in or learned expectations enable them to learn certain procedures more easily than others*. For example, it is easier for us to learn that a burst of loud noise means danger than to learn that a particular melody means danger. Also, we are accustomed to turning a knob clockwise to increase the volume on a radio or television, so we expect to turn the knob clockwise on any new machine to increase almost anything. Similarly, we expect to turn the knob clockwise to move something to the right and counterclockwise to move something to the left (as we do when steering a car). These examples may seem obvious, but in many cases ergonomists must do trial-and-error research to find the best way to set up the controls for a new device.

At some universities, the ergonomics program is part of the psychology department; at others it is part of engineering; at still others, it is administered jointly by both departments. Regardless of who administers the program, ergonomics necessarily combines features of psychology, engineering, and computer science.

School Psychology A great many children have academic problems at one time or another. Some children simply have trouble sitting still or paying attention. Others get into trouble for misbehavior. Others have specialized problems with reading, spelling, arithmetic, or other academic skills. Some have distractions in their home life that prevent them from concentrating on their studies. Other children, because of their special gifts or talents, master their schoolwork faster than most children and become bored. They, too, need special attention.

School psychologists are *specialists in the psychological condition of students*, usually in kindergarten

through 12th grade. The role of school psychologists varies substantially, depending on the psychologists themselves, their education, and the school system for which they work. Broadly speaking, school psychologists identify the educational needs of children, devise a plan to meet those needs, and then either implement the plan themselves or advise teachers how to implement it. As of 1992, there were an estimated 87,000 active school psychologists worldwide (Oakland & Cunningham, 1992).

School psychology can be taught in a psychology department, a branch of the department of education, or a special department of educational psychology. In some countries, it is possible to practice school psychology with only a bachelor's degree. In the United States, the minimum is usually a master's degree, but job opportunities are much greater for people with a doctorate degree, and in the future, a doctorate may become necessary. The number of people employed as school psychologists has been growing sharply, and job opportunities are expected to continue to be plentiful. Most school psychologists work for a school system; others work for mental health clinics, guidance centers, and other institutions.

Table 1.2 summarizes some of the major fields of psychology, including several not discussed in the previous sections.

Should You Major in Psychology?

Can you get a job if you major in psychology? Psychology is one of the most popular majors in the United States, Canada, and Europe. So, if psychology majors cannot get jobs, a huge number of people are going to be in trouble!

The bad news is that very few jobs specifically advertise for college graduates with a bachelor's degree in psychology. The good news is that an enormous variety of jobs are available for graduates with a bachelor's degree, not specifying any major. Therefore, if you get a degree in psychology, you will compete with history majors, English majors, astronomy majors, physics majors, and lots of others for jobs in government, business, and industry. Some of those jobs do have direct relevance to psychology, particularly in personnel work or social services. Even if you get a job that seems remote from psychology, your psychology courses will have taught you a good deal about how to evaluate evidence, organize and write papers, handle statistics, listen carefully to what people say, understand and respect cultural differences, and so forth. In short, a psychology major can provide a useful background for a variety of jobs.

Psychology also provides a good background for people entering professional schools. Many students major in psychology and then apply to medical school, law school, divinity school, or other programs. Find out what course work is expected for the professional program of your choice, and

TABLE 1.2 Some Major Specializations in Psychology

SPECIALIZATION	GENERAL INTEREST	EXAMPLE OF SPECIFIC INTEREST OR RESEARCH TOPIC
<i>Clinical psychologist</i>	Emotional difficulties	How can people be helped to overcome severe anxiety?
<i>Community psychologist</i>	Organizations and social structures	Would improved job opportunities decrease certain types of psychological distress?
<i>Counseling psychologist</i>	Helping people to make important decisions and to achieve their potential	Should this person consider changing careers?
<i>Developmental psychologist</i>	Changes in behavior as people grow older	At what age can a child first distinguish between appearance and reality?
<i>Educational psychologist</i>	Improvement of learning in school	What is the best way to test a student's knowledge?
<i>Environmental psychologist</i>	The influence of noise, heat, crowding, and other environmental conditions on human behavior	How can a building be designed to maximize the comfort and productivity of the people who use it?
<i>Ergonomist</i>	Communication between person and machine	How can an airplane cockpit be redesigned to increase safety?
<i>Experimental psychologist</i>	Sensation, perception, learning, thinking, memory	Do people have several kinds of learning? Do they have several kinds of memory?
<i>Industrial and organizational psychologist</i>	People at work, production efficiency	Should jobs be made simple and foolproof or interesting and challenging?
<i>Personality researcher</i>	Personality differences	Why are certain people shy and others gregarious?
<i>Biopsychologist</i>	Relationship between brain and behavior	What body signals indicate hunger and satiety?
<i>Psychometrician</i>	Measurement of intelligence, personality, and interests	How fair are current IQ tests? Can we devise better tests?
<i>School psychologist</i>	Problems that affect schoolchildren	How should the school handle a child who regularly disrupts the classroom?
<i>Social psychologist</i>	Group behavior, social influences	What methods of persuasion are most effective for changing attitudes?

then compare the course work required for a psychology major. You will probably find considerable overlap.

If you want a career in psychology, you can greatly increase your options with an advanced degree, especially a doctorate. A doctorate will qualify you to apply for positions as a college professor or, depending on your area of specialization, jobs in hospitals, clinics, private practice, school systems, business and industry. If you are a first- or second-year college student now, it would probably take you at least eight years and probably longer, to get a doctorate, and no one can accurately predict the job market that far into the future. Learn as much as you can, both inside and outside the field of psychology, and keep your options open.

The types of people majoring in psychology have become more diverse over the years. For a long time, academic psychology, like most other academic disciplines, was dominated by men. Women students were not encouraged to seek a Ph.D. degree; those who did were rarely offered professorships at the most prestigious colleges or universities. Since the early 1970s, though, women have received a greatly increasing percentage of the advanced degrees in almost all academic fields. Figure 1.5 shows the growing percentage of women in psychology and several other fields (Pion et al., 1996). Women now receive more than half of the new doctorates in psychology, not only in the United

States but in all European countries as well (Newstead & Makinen, 1997), and women dominate psychology more than most other academic and professional fields. Figure 1.6 shows the increasing representation of women separately in several subdivisions of psychology. Note the strong dominance in developmental psychology.

Minorities also constitute a growing percentage of psychologists, although the total number is still small. According to a 1988 survey in the United States, African-Americans, Hispanics, Asian-Americans, and other ethnic minorities combined received 11.4% of the doctorates awarded in clinical psychology. That percentage is low compared to many other academic fields. Many graduate schools now actively seek applications from minority students who would like to become psychologists (Hammond & Yung, 1993).

THE MESSAGE

Types of Psychologists

An experimental psychology researcher, a clinical psychologist, an ergonomist, and an industrial/organizational psychologist are all psychologists, even though their daily

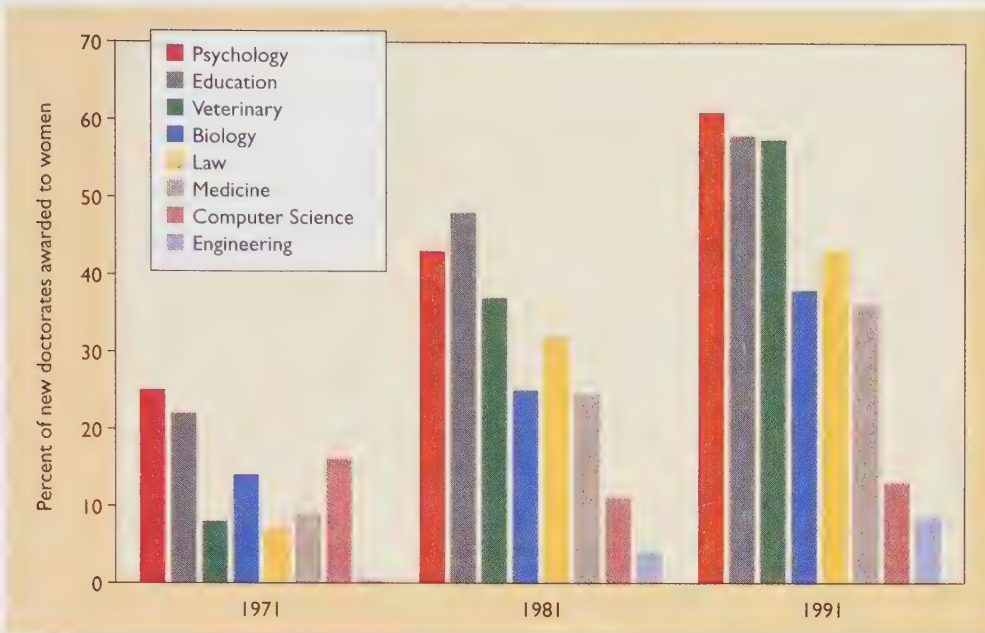


FIGURE 1.5 Between 1971 and 1991, the percentage of new doctorate degrees awarded to women in the United States increased dramatically in most fields. In 1991, women received more than half of the new doctorate degrees in psychology, education, and veterinary medicine. (Based on data of Pion et al., 1996.)

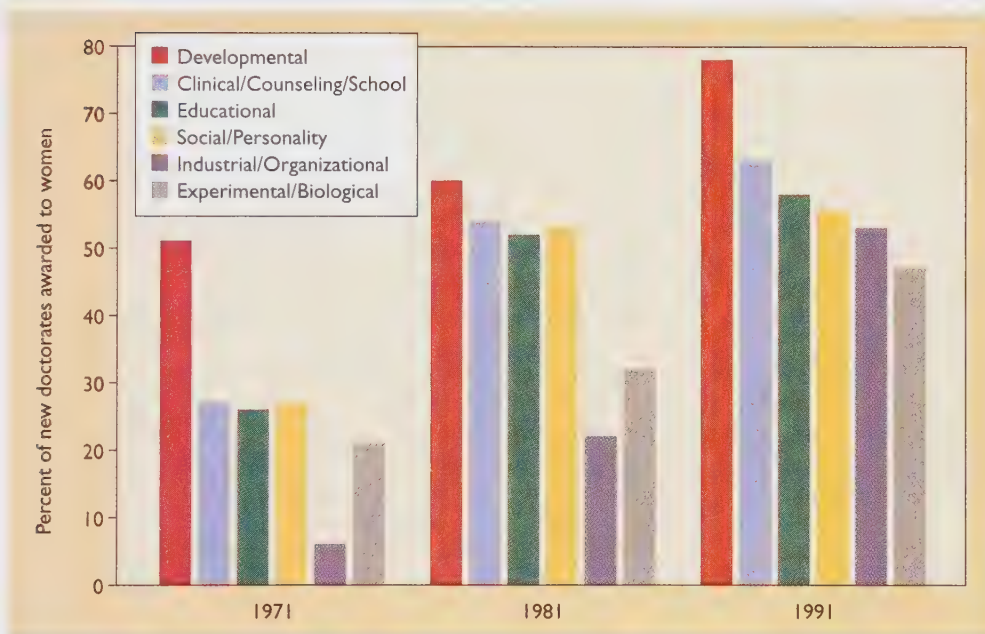


FIGURE 1.6 Women now dominate in some fields of psychology, such as developmental, more than in other fields, such as experimental and biological, although the percentage of women has increased in all of these fields. (Based on data of Pion et al., 1996.)

activities have little in common. What does unite psychologists is a dedication to psychological progress through research.

I have oversimplified this discussion of the various psychological approaches in several ways. First, it is only partly correct to refer to biological psychology, cognitive psychology, social psychology, and other fields as approaches. True, each constitutes one way of approaching certain phenomena of interest to all psychologists. But each is also a separate field of study with its own special phenomena. Biological psychologists ask questions about how the brain works; social psychologists ask questions about group behavior. Furthermore, the various approaches overlap significantly.

Nearly all psychologists combine insights and information gained from a variety of approaches. To understand why one person differs from another, most psychologists are interested in their biology, their past learning experiences, the social influences that have affected them, and much more.

As we proceed through this book, we shall consider one type of behavior at a time and, generally, one approach at a time. That is simply a necessity; we cannot talk intelligently about all kinds of psychological processes at once. But bear in mind that all these processes do ultimately fit together; what you do at any given moment depends on your biology, your past experiences, your social setting, your emotions, and a great deal more.

S U M M A R Y *

- ★ **What is psychology?** Psychology is the systematic study of behavior and experience. Psychologists deal with both theoretical and practical questions. (page 5)
- ★ **Determinism/free will.** Determinism is the view that everything that occurs, including human behavior, has a physical cause. That view is difficult to reconcile with the conviction that humans have free will—that we deliberately, consciously decide what to do. (page 5)
- ★ **Mind-brain.** The mind-brain problem is the question of how conscious experience is related to the activity of the brain. (page 6)
- ★ **Nature-nurture.** Behavior depends on both nature (heredity) and nurture (environment). Psychologists try to determine the influence of those two factors on differences in behavior. The relative contributions of nature and nurture vary from one behavior to another. (page 8)
- ★ **Research fields in psychology.** Psychology is an academic field with many subfields, including biological psychology, learning and motivation, cognitive psychology, developmental psychology, and social psychology. (page 9)
- ★ **Psychology versus psychiatry.** Clinical psychologists have either a Ph.D., Psy.D., or master's degree; psychiatrists are medical doctors. Both clinical psychologists and psychiatrists treat people with emotional problems, but only psychiatrists can prescribe drugs and other medical treatments. (page 11)
- ★ **Nonclinical applied psychology.** Nonclinical fields of application include industrial/organizational psychology, ergonomics, and school psychology. (page 11)
- ★ **Job prospects.** People with a bachelor's degree in psychology enter a wide variety of careers or continue their education in professional schools. Those with a doctorate in psychology have additional possibilities, depending on their area of specialization. In psychology, as in any other field, job prospects can change between the start and finish of one's education. (page 13)

Suggestion for Further Reading

Sechenov, I. (1965). *Reflexes of the brain*. Cambridge, MA: MIT Press. (Original work published 1863.) One of the first attempts to deal with behavior scientifically and still one of the clearest and most stimulating statements of the argument for determinism in human behavior.

Terms

- psychology** the systematic study of behavior and experience (page 5)
- determinism** the assumption that all behavior has a cause, or *determinant*, in the observable world (page 5)

- free will** the doctrine that behavior is caused by a person's independent decisions, not by external determinants (page 7)
- mind-brain problem** the philosophical question of how the conscious mind is related to the physical nervous system, including the brain (page 8)
- nature-nurture issue** the question of the relative roles played by heredity (nature) and environment (nurture) in determining differences in behavior (page 9)
- biopsychologist (or behavioral neuroscientist)** a specialist who tries to explain behavior in terms of biological factors, such as electrical and chemical activities in the nervous system, the effects of drugs and hormones, genetics, and evolutionary pressures (page 9)
- learning and motivation** study of how behavior depends on the outcomes of past behaviors and on current motivations (page 9)
- cognition** thinking and acquiring knowledge (page 9)
- cognitive psychologist** a specialist who studies thought processes and the acquisition of knowledge (page 9)
- developmental psychologist** a specialist who studies the behavioral capacities of different ages and how behavior changes with age (page 10)
- social psychologist** a specialist who studies how an individual influences others and is influenced by other people (page 10)
- psychotherapist** a specialist who provides help for people with psychological problems (page 10)
- clinical psychologist** someone with an advanced degree in psychology, with a specialization in understanding and helping people with psychological problems (page 11)
- psychiatry** a branch of medicine that deals with emotional disturbances (page 11)
- psychoanalyst** a psychotherapist who relies heavily on the theories of Sigmund Freud (page 11)
- industrial/organizational psychology** the study of people at work (page 11)
- ergonomist or human factors specialist** a psychologist with engineering skills who works to facilitate the operation of machinery so that the average user can use it as efficiently and as safely as possible (page 13)
- principle of compatibility** the concept that people's built-in or learned expectations enable them to learn certain procedures more easily than others (page 13)
- school psychologist** a specialist in the psychological condition of students (page 13)

Answers to Concept Checks

1. **a.** Developmental psychology. **b.** Social psychology. **c.** Cognitive psychology. **d.** Biological psychology. **e.** Learning and motivation. (page 10)
2. Most psychoanalysts can prescribe drugs, because most are psychiatrists, and psychiatrists are medical doctors. However, psychoanalysts who are psychologists are not medical doctors and therefore cannot prescribe drugs. (page 11)

*The page numbers following each item indicate where you can review a topic.

Web Resources

What Is Psychology?

www.kean.edu/~psych/What_is.html

The Psychology Department at Kean University offers information about the discipline of psychology, some of the major subfields of psychology, and careers available to holders of undergraduate, master's, and doctoral degrees.

Psychological Science Agenda

www.apa.org/psa/

Psychological Science Agenda, the newsletter of the American Psychological Association Science Directorate, carried a series of articles written by people working in interesting

and unusual nonacademic careers. Among the careers: Research in the Public Sector, Highway Safety Research, Market Research and Consulting, Acquisition and Sponsoring Editor, Director of Education and Research, Human Factors and User Interface Design, Trial Consultant, and Executive Search Consultant. The APA Science Directorate now sponsors a Personal Stories of Nonacademic Careers Web page (www.apa.org/science/psa.html) that lists most of these, and others.

MODULE 1.2

Psychology Then and Now

How did psychology originate?

How is it different now from what it was in its
earliest era?

Imagine yourself as a young scholar in about 1880. Enthusiastic about the new scientific approach in psychology, you have decided to become a psychologist yourself. Like other early psychologists, you have a background in either biology or philosophy. You are determined to apply the scientific methods of biology to the problems of philosophy.

So far, so good. But what questions will you address? A good, worthwhile research question is both important and answerable. In 1880, how would you decide which questions are important? You cannot get research ideas from a psychological journal, because the first issue won't be published until next year. You cannot follow in the tradition of previous researchers, because there haven't *been* any previous researchers. You are on your own.

Furthermore, in the late 1800s and early 1900s, psychologists were not yet sure which questions were answerable. Sometimes they are still unsure: Should we try to study the nature of human consciousness, or should we skip conscious experience and instead concentrate on describing what people actually do? Many of the changes that have occurred during the history of psychology have been changes in investigators' decisions about what constitutes a good research question.

In the next several pages, we shall explore some of these changes in psychological research questions, featuring a few projects that dominated psychology for a while and then faded from interest. I shall not mention Sigmund Freud or several other early pioneers whose contributions will be discussed in later chapters.

Psychology in the Early Era (1879 to about 1920)

At least since Aristotle (384–322 B.C.), philosophers have been debating why people act the way they do, why they have the experiences they do, and why one person is different from

another. Writers such as Chaucer, Dostoyevsky, and Goethe have also made profound observations on human behavior.

Without discounting the importance of these great thinkers, several 19th-century scholars wondered whether a scientific approach would be fruitful. They were impressed by the great strides made in physics, chemistry, and biology; they believed that similar progress could be made in psychology if evidence were collected and evaluated scientifically.

Wilhelm Wundt and the First Psychological Laboratory

The origin of psychology, as we now know it, is generally dated to 1879, when a medical doctor and sensory researcher named Wilhelm Wundt (pronounced “voont”) set up the first psychology laboratory, in Leipzig, Germany. Wundt and others had conducted psychological experiments before, but this was the first time anyone had established a laboratory exclusively for psychological research.

Wundt's fundamental question was: What are the components of experience, or mind? He proposed that psychological experience is composed of compounds, just as chemistry has compounds. Psychology, he maintained, has two kinds of elements—sensations and feelings (Wundt, 1896/1902)¹. So, at any particular moment you might experience the taste of a fine meal, the sound of good music, and a certain degree of pleasure. These would merge together into a single experience, but that experience would still include the separate elements. Furthermore, Wundt maintained, your experience is partly under your control; even when the physical situation stays the same, you can shift your attention from one element to another and get a different experience.

Wundt's question about the components of experience was a philosophical one, and some of his opinions about the elements of the mind resembled the writings of philosophers before him. But Wundt, unlike these philosophers, tried to test his statements by collecting data. He presented various kinds of lights, textures, and sounds and asked subjects to report the intensity and quality of their sensations. He measured the changes in people's experiences as he changed the stimuli.

Wundt also demonstrated that it was possible to conduct meaningful psychological experiments. For example, in one of Wundt's earliest studies, he set up a pendulum that struck metal balls and made a sound at two points on its swing (points b and d in Figure 1.7). He or another subject would watch the pendulum and determine where it appeared to be when they heard the sound. In some cases, the pendulum appeared to be slightly in front of the ball and, in other cases, slightly behind the ball. The apparent position of the pendulum at the time of the sound differed from its actual position by an average of $\frac{1}{8}$ of a second (Wundt, 1862/1961). Apparently, we can be slightly wrong about the time that we see or hear something. Wundt's interpretation

¹ A reference containing a slash, such as this one, refers to a book originally published in the first year (1896) and reprinted in the second year (1902).

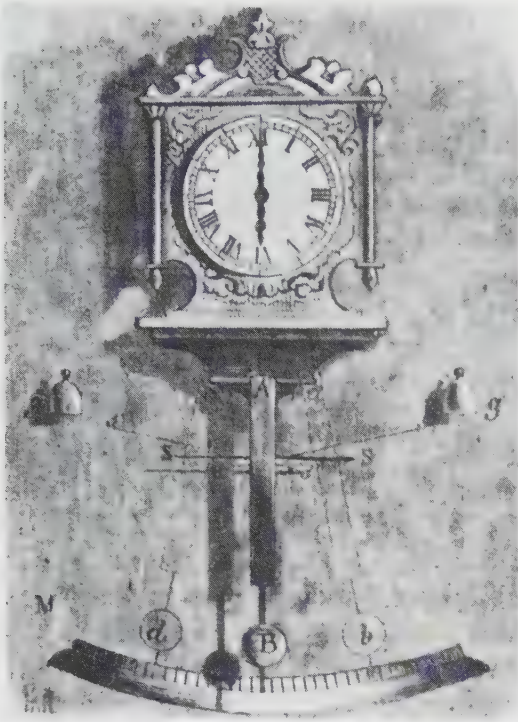


FIGURE 1.7 (Left) In one of Wilhelm Wundt's earliest experiments, the pendulum struck the metal balls (b and d), making a sound each time. To an observer, however, the ball appeared to be somewhere else at the time of the sound, generally the distance that it would travel in about $\frac{1}{8}$ of a second. Wundt inferred that a person needs about $\frac{1}{8}$ of a second to shift attention from one stimulus to another. (Right) The Walt Disney studios rediscovered Wundt's observation decades later: The character's mouth movements seem to be in synchrony with the sounds if the movements precede the sounds by $\frac{1}{8}$ to $\frac{1}{6}$ of a second.

was that a person needs about $\frac{1}{8}$ second to shift attention from one stimulus to another.

Wundt and his students were prolific investigators; the brief treatment here does not do him justice. He contributed a great deal (writing more than 50,000 pages), but his most lasting impact on the field of psychology came from setting the precedent of studying psychological questions by collecting scientific data.

Edward Titchener and Structuralism

For years, most of the world's scientific psychologists received their education from Wilhelm Wundt himself. One of Wundt's students, Edward Titchener, came to the United States in 1892 as a psychology professor at Cornell University. Like Wundt, Titchener believed that the main question of psychology was the nature of mental experiences.

Titchener (1910) typically presented a stimulus and asked his subject to analyze it into its separate features—for example, to look at an apple and describe its redness, its brightness, its shape, and so forth. He called his approach **structuralism**, *an attempt to describe the structures that compose the mind*. He was less interested in what those elements *do* (their functions).

If you asked psychologists today whether they thought Titchener had correctly described the structures of the mind, you would probably get blank looks or shrugs of the shoulders. After Titchener died in 1927, psychologists virtually abandoned his research methods and even the questions he asked. Why? Remember that a good scientific question is both important and answerable. Regardless of whether Titchener's question about the elements of the mind was important, after all of his efforts, the question

seemed less and less answerable. The main reason was that, when observers described their experiences, Titchener could not check the accuracy of their descriptions.

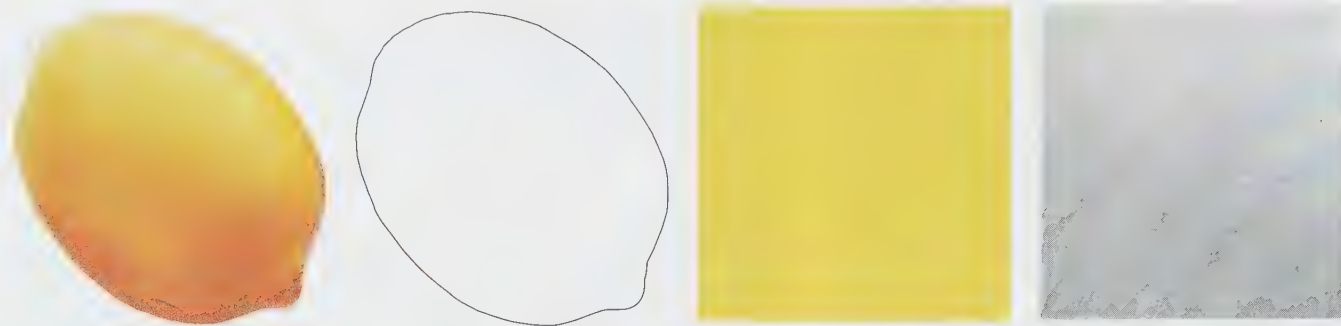
For example, suppose you are the psychologist: I look at a lemon and tell you I have an experience of brightness that is totally separate from my experience of yellowness. How do you know whether I am lying to you, telling you what I think you want me to say, or even deceiving myself? You must take my word for it, unless you have another way of seeing into my mind. Other psychologists' frustration with this approach eventually turned many of them against studying the mind and toward studying observable behaviors.

William James and Functionalism

Almost simultaneously with the work of Wundt and overlapping the work of Titchener, Harvard University's William James articulated some of the major issues of psychology and won eventual recognition as the founder of American psychology. James's book *The Principles of Psychology* (James, 1890) defined the questions that dominated psychology for years afterward, and even to some extent today.

James had little patience for either Wundt's or Titchener's search for the elements of the mind. He focused concern on the actions that the mind *performs*, rather than the ideas that the mind *has*. That is, he did not care to isolate the elements of consciousness; he preferred *to learn how the mind produces useful behaviors*. For this reason, we call his approach **functionalism**. He suggested the following examples of good psychological questions (James, 1890):

- How can people strengthen good habits?
- How many objects can a person attend to at once?



Edward Titchener asked subjects to describe their sensations. For example, they might describe their sensation of shape, their sensation of color, and their sensation of texture while looking at a lemon. Titchener had no way to check the accuracy of these reports, however, so later psychologists abandoned his methods.

- How do people recognize that they have seen something before?
- How does an intention lead to an action?

James proposed some possible answers, but did little research of his own to address these questions. His main contribution was to inspire later researchers to address the questions that he posed.

Studies of Sensation

For many early psychologists, the most important question was the relationship between physical stimuli and psychological sensations. To a large extent, the study of sensation *was* psychology. The first English-language textbook of the “new,” scientifically based psychology devoted almost half of its pages to a discussion of the senses and the related topic of attention (Scripture, 1907). By the 1930s, standard psychology textbooks devoted less than 20% of their pages to these topics (Woodworth, 1934); today, the coverage of this topic constitutes about 5–10%.

Why were early psychologists so interested in sensation? One reason was philosophical: They wanted to understand mental experience, and experience is composed mostly if not entirely of sensations. The other reason was strategic: If psychologists were to demonstrate the possibil-

ity of a scientific psychology, they needed to begin with questions that were answerable, and many questions about sensation are indeed answerable.

Early psychologists discovered that what we see, hear, and otherwise sense is not the same as what is actually there. For example, the *perceived* intensity of a stimulus is not directly proportional to the *actual* physical intensity of the stimulus: For example, a light that is twice as intense as another light does not look twice as bright. Figure 1.8 shows the actual relationship between the intensity of light and its perceived brightness. *The mathematical description of the relationship between the physical properties of a stimulus and its perceived properties* is called the **psychophysical function**, because it relates psychology to physics. Such research demonstrated that, at least in the study of sensation, scientific methods can provide nonobvious answers to psychological questions.

The Influence of Darwin and the Study of Animal Intelligence

Charles Darwin's theory of evolution by natural selection (Darwin, 1859, 1871) had an enormous impact not only on biology but also on psychology. Darwin argued that humans and other species share a remote common ancestor. This

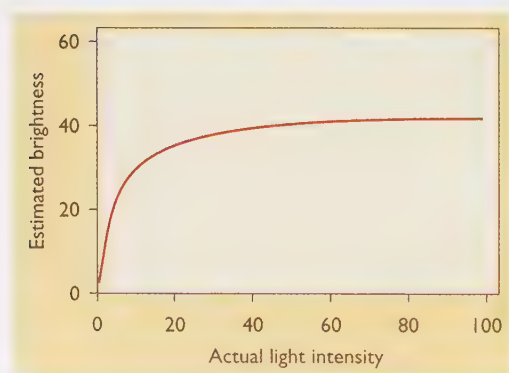


FIGURE 1.8 This graph of a psychophysical event shows the perceived intensity of light versus its physical intensity. When a light becomes twice as intense physically, it does not seem twice as bright. (Adapted from Stevens, 1961.)

proposal implied that each species had evolved specializations adapted to different ways of life, but it also implied that all vertebrate species had certain basic features in common. It implied also that nonhuman animals should exhibit varying degrees of human characteristics, including intelligence similar to human intelligence.

Presuming that we accept this last implication, what should psychologists do about it? Some early **comparative psychologists**, *specialists who compare different animal species*, did something that seemed reasonable at first, though it later seemed less and less fruitful: They set out to measure animal intelligence. They apparently imagined that they would be able to rank-order all animals from the smartest to the dumbest. Toward that goal, they set various species to such tasks as the delayed-response problem and the detour problem. In the *delayed-response problem*, an animal was given a signal indicating where it could find food. Then the animal was delayed in its movement toward the food (Figure 1.9), to find out how long each species could remember the signal. In the *detour problem*, an animal was separated from food by a barrier (Figure 1.10), to find out which species would take a detour away from the food at first in order to get to it later.

Comparative psychologists set their animals to other tasks as well (Maier & Schneirla, 1964). But measuring animal intelligence turned out to be much more difficult than it sounded. Too often a species that seemed dull-witted with one task seemed highly capable on a very similar one. For example, zebras are generally slow to learn to approach one pattern instead of another for food, unless the patterns happen to be narrow stripes versus wide stripes, in which case zebras become geniuses (Giebel, 1958). (See Figure 1.11.) Rats perform very poorly when they have to find an object that looks different from the others but very well when they have to find an object that smells different (Langworthy & Jennings, 1972).

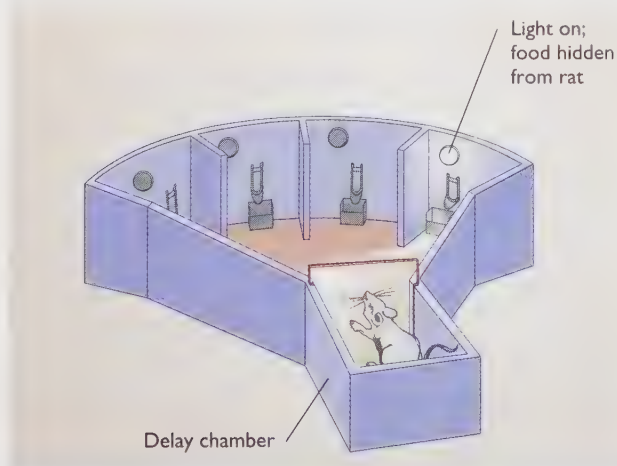


FIGURE 1.9 One of the tasks used by early comparative psychologists to assess animal intelligence tested the delayed-response problem. A stimulus was presented, and a delay ensued; then the animal was expected to respond to the remembered stimulus. Variations on this delayed-response task are still used today.

Eventually, psychologists realized that the relative intelligence of nonhuman animals was a complicated and perhaps meaningless question. Some researchers today even hesitate to say that monkeys are more intelligent than frogs (Macphail, 1985), preferring instead to talk about which particular sensory or learning functions are better developed in one species or another.

The Measurement of Human Intelligence

Some psychologists studied animal intelligence, but most had an even greater interest in human intelligence. Francis Galton, a first cousin of Charles Darwin, was among the

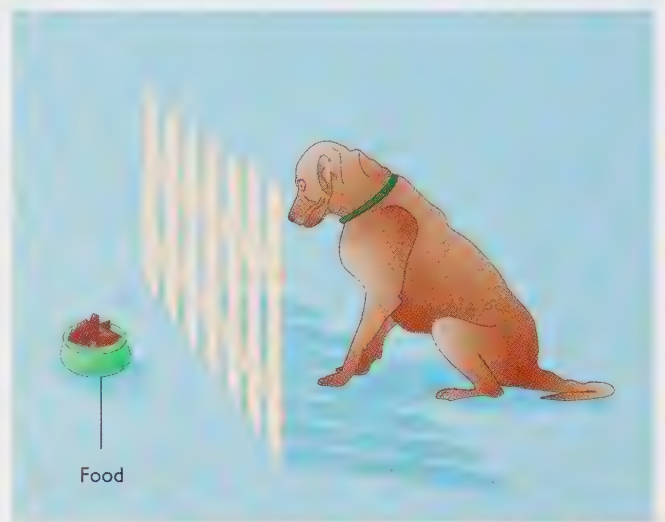


FIGURE 1.10 Another task popular among early comparative psychologists was the detour problem. An animal needed to first go away from the food in order to move toward it.



FIGURE 1.11 Zebras learn rapidly when they have to compare stripe patterns (Giebel, 1958). How "smart" a species is perceived to be depends in part on what ability or skill is being tested.

first to try to measure intelligence and to ask whether intellectual variations were based on heredity. Galton was fascinated with the measurement of almost anything and pioneered many new methods of measurements (Hergenhahn, 1992). For example, he invented the weather map, measured degrees of boredom during lectures, suggested the use of fingerprints to identify individuals, and—presumably, in the name of science—attempted to measure the degree of beauty of women in different countries.

In an effort to determine the role of heredity in human achievement, Galton (1869/1978) examined whether the sons of famous and accomplished men were likely to become equally eminent themselves. (He paid little attention to women, on the grounds that women in 19th-century England had little prospect of attaining eminence.) Galton found that the sons of judges, writers, politicians, and other noted men had a high probability of reaching high levels of accomplishment themselves. He attributed this edge to heredity, although today we would hardly consider his evidence persuasive: Sons of eminent men clearly had a favorable environment, not just favorable genes. Furthermore, he suggested that eminence is partly due to intelligence; thus, he believed that a tendency toward high or low intelligence was heritable.

Galton, however, had no valid test of intelligence. He attempted to measure intelligence using simple tests of sensory capacities and motor coordination, but none of these tests proved satisfactory. In 1905 a French researcher, Alfred Binet, devised the first useful intelligence test, which we shall discuss further in Chapter 9. At this point, we will just note that the idea of testing intelligence quickly captured a great deal of interest in the United States and other Western countries. Psychologists, perhaps inspired by the popularity of intelligence tests, later developed tests of personality, interests, and other psychological characteristics.

The Role of Women in the Early Days of Psychology

In the late 1800s and early 1900s, most U.S. colleges provided very limited opportunities for women, either as students or as faculty. Psychology was not much different from other fields in this regard, but a few women did make major contributions and achieved prominence.

One of the first to do so was Mary Calkins (Scarborough & Furomoto, 1987). When Henry Durant founded Wellesley College in 1870, he decided to hire only women to teach the all-female student body, but he could find no women with an advanced degree in psychology. Finally, in 1890, he hired a bright young woman, Mary Calkins (Figure 1.12), who had a B.A. degree in classics, to teach psychology, promising to pay for her graduate education in psychology. Then, he needed to find a graduate program that would accept a female student. After much debate and resistance, nearby Harvard University finally agreed to let her attend graduate classes. In 1895, when she passed the final examination for the Ph.D. degree, one of her professors re-



FIGURE 1.12 Mary Calkins, one of the first prominent women in U.S. psychology.

marked that she had performed better on the exam than had any other student in the history of the department.

The Harvard administration, however, was still unwilling to grant a Ph.D. degree to a woman and suggested a compromise instead: Calkins would receive a Ph.D. degree from Radcliffe College, the recently established women's undergraduate college associated with Harvard. Calkins refused, declaring that to accept the compromise would violate the high ideals of education. She never gave in, and neither did Harvard. Although Mary Calkins never did receive a Ph.D. degree, she became a pioneer in psychological research, served as president of the American Psychological Association, and invented a technique for studying memory, known as the paired-associates method, that is still used today.

The first woman to receive a Ph.D. degree in psychology was Margaret Washburn, who received her degree from Cornell University in 1894. She later wrote *The Animal Mind* (1908), the first textbook on that topic; she also served as president of the American Psychological Association. Christine Ladd-Franklin, another early psychologist, did outstanding research on vision, beginning in 1887.

The Rise of Behaviorism

Earlier in this chapter, I casually defined psychology as “the systematic study of behavior and experience.” Most psychologists today would probably offer a similar definition, but for a substantial period of psychology's history, most experimental psychologists would have objected to the words “and experience.” Some psychologists still object today, a little less strenuously. From about 1920 to 1960 or 1970, most laboratory researchers described psychology as simply the study of behavior, period. These researchers had little to say about minds, experiences, or anything of the sort. (According to one quip, psychologists had “lost their minds.”)

How did psychologists reach this conclusion? Recall that Titchener's effort to analyze experience into its components had failed. Most psychologists then concluded that questions about the mind were unanswerable. The comparative psychologists had also failed in their efforts to

measure animal intelligence, but in the process they had developed techniques for studying animal learning. Many psychologists discarded the question about animal intelligence but kept the research methods. The research questions they addressed were broad and simple: What do people and other animals do? And how do changes in the environment or in their experience change what they do? These questions were clearly meaningful, although no one knew yet whether the answers would consist of long lists of details or a short list of general laws.

John B. Watson and the Origin of Behaviorism

We can regard John B. Watson as the founder of **behaviorism**, *a field of psychology that concentrates on observable, measurable behaviors and not on mental processes*. Watson was not the first behaviorist—actually, it is difficult to say who was the first—but he systematized the approach, popularized it, and stated its goals and assumptions (Watson, 1919, 1925). Here are two quotes from Watson:

Psychology as the behaviorist views it is a purely objective experimental branch of natural science. Its theoretical goal is the prediction and control of behavior. (Watson, 1913, p. 158)

The goal of psychological study is the ascertaining of such data and laws that, given the stimulus, psychology can predict what the response will be; or, on the other hand, given the response, it can specify the nature of the effective stimulus. (Watson, 1919, p. 10)

Studies of Learning

Inspired by Watson, many researchers set out to study animal behavior, especially animal learning. Many optimistically expected to discover simple, basic laws of behavior

comparable to Newton's laws of physics. Many experimenters believed that behavioral laws would be more or less the same from one species to another, so they studied animals, especially rats. By 1950, one psychologist found that well over half of all psychological studies on animals used rats as subjects (Beach, 1950).

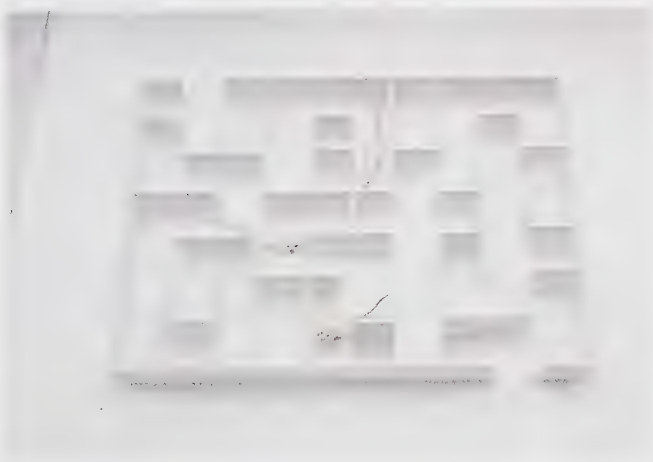
Researchers simplified their task by not only concentrating on a few convenient species, but also by focusing on a few convenient examples of learning, such as maze learning. One highly influential Yale psychologist, Clark Hull, was very explicit about what he considered an important psychological question: "One of the most persistently baffling problems which confronts modern psychologists is the finding of an adequate explanation of the phenomena of maze learning" (Hull, 1932). Would you list that topic today as one of the most important things you would like to understand in psychology?

As research progressed, however, psychologists found that even the behavior of a rat in a maze was far more complicated than they had expected. As a result, they largely lost interest in maze studies, so the kinds of maze studies that were popular in the 1950s are a rarity today. Just as psychologists of the 1920s decided that Titchener's structuralist approach was unlikely to answer his questions within a reasonable period of time, psychologists of a later era abandoned the hope of discovering everything important about psychology by studying a few simple learning situations.

The Current Era in Psychology

The rest of this book will focus on the current era in psychology, with occasional flashbacks to the history of particular subfields. Psychology today is an extremely diverse field. Some researchers are trying to isolate simple mechanisms of behavior; others are attempting to understand highly complex processes; still others engage in a wide variety of practical applications.

Psychologists today have also broadened their scope to include more of human diversity. In an earlier era, psychological researchers who were seeking general laws or principles of behavior often assumed that they could discover these by studying rats, pigeons, or any other convenient species. When they studied humans, they generally limited themselves to convenient samples, such as college students. That strategy is justifiable for certain purposes—such as understanding how the eyes work—including characteristics that do not vary from one group of people (or animals) to another. For other purposes, the differences among people may be more important than the similarities, however. For example, when studying child-rearing practices, the growth of self-esteem, or marriage customs, psychologists compare the results from different cultures throughout the world and of various ethnic groups within a



Early behaviorists studied rats in mazes, hoping to find general laws of behavior. As they discovered that this apparently simple behavior was very complicated, their interest declined and they turned to other topics.

Social psychologists study how our behavior depends on cultural influences, the expectations of others, and what we observe in other people's behavior.



given country. **Cross-cultural studies**, *research that compares people from various cultures*, have become increasingly influential in psychology.

Another noteworthy change in psychology is the explosive growth of neuroscience (the study of the brain). At one time, most psychologists thought they could safely ignore brain research, but no more. A theoretical understanding of behavior requires an understanding of the brain mechanisms behind it, and you will find references to biology in many sections of this text.

THE MESSAGE

Psychology Through the Years

Throughout the early years of psychology, many psychologists went down blind alleys, devoting enormous efforts to projects that produced disappointing results. Not all the efforts of early psychologists were fruitless, though; in later chapters, you will encounter some classic studies that have withstood the test of time. Still, if psychologists of the past spent countless years on fashionable projects, only to decide later that their efforts were misguided, how do we know that many psychologists aren't on the wrong track right now?

The answer is that we don't. Thousands of psychologists are engaged in various kinds of research, and chances are, many of them are working on projects that will never accomplish much. As you read through later chapters, you are welcome to entertain doubts. Maybe some psychologists' questions are not so simple as they seem; perhaps some of their answers are not very solid; perhaps you can think of a better way to approach certain topics.

In short, psychologists do not have all the answers. But that is not a reason for despair. Much like a rat in a maze, researchers make progress by trial and error. They pose a question, try a particular research method, and find out what happens. Sometimes the results are fascinating and rich in practical consequences; sometimes they turn out to be puzzling or inconclusive. If one study after another proves disappointing, psychologists either look for a new method or else they change the question they are asking. By abandoning unsuccessful approaches, they eventually find the way to better questions and better answers.

SUMMARY

- ★ **Choice of research questions.** During the history of psychology, researchers have several times changed their opinions about what constitutes an interesting, important, answerable question. (page 18)
- ★ **First experiments.** In 1879 Wilhelm Wundt established the first laboratory devoted to psychological research. He demonstrated the possibility of psychological experimentation. (page 18)
- ★ **Limits of self-observation.** One of Wundt's students, Edward Titchener, attempted to analyze the elements of mental experience, relying on people's own observations. Other psychologists became discouraged with this approach. (page 19)
- ★ **The founding of American psychology.** William James, generally considered the founder of American psychology, focused attention on how the mind guides useful behavior, rather than on the contents of the mind. By doing so, James paved the way for the rise of behaviorism. (page 19)

★ **Early sensory research.** In the early days of psychology, many researchers concentrated on studies of the senses, partly because they were more likely to find definite answers on this topic than on other topics. (page 20)

★ **Darwin's influence.** Charles Darwin's theory of evolution by natural selection influenced psychology in many ways; it prompted some prominent early psychologists to compare the intelligence of different species. That question turned out to be more complicated than anyone had expected. (page 20)

★ **Intelligence testing.** The measurement of human intelligence was one concern of early psychologists that has persisted through the years. (page 21)

★ **Women in the early days of psychology.** In spite of an environment that discouraged women from pursuing academic careers, several women (including Mary Calkins and Margaret Washburn) became leaders during the early days of psychology. (page 22)

★ **The era of behaviorist dominance.** As psychologists became discouraged with their attempts to analyze the mind, they turned to behaviorism. For many years, psychological researchers concentrated on behavior, especially animal learning, to the virtual exclusion of mental experience. (page 22)

★ **Maze learning.** Clark Hull exerted great influence for a number of years. Eventually, his approach became less popular because rats in mazes did not seem to generate simple or general answers to major questions. (page 23)

★ **Psychological research today.** Today, psychologists study a wide variety of topics. Still, we cannot be certain that we are not currently going down some blind alleys, just as many psychologists did before us. (page 23)

Suggestion for Further Reading

Scarborough, E., & Furomoto, L. (1987). *Untold lives: The first generation of American women psychologists*. New York: Columbia University Press. A rich account of history and biography.

Terms

structuralism an attempt to describe the structures that compose the mind (page 19)

functionalism an attempt to understand how mental processes produce useful behaviors (page 19)

psychophysical function the mathematical description of the relationship between the physical properties of a stimulus and its perceived properties (page 20)

comparative psychologist a specialist who compares different animal species (page 21)

behaviorism a field of psychology that concentrates on observable, measurable behaviors and not on mental processes (page 23)

cross-cultural studies research studies that compare people from various cultures (page 24)

Web Resources

HistPsyc Headlines Pages

www.unb.ca/web/units/psych/likely/headlines/

David Likely, at the University of New Brunswick, produced the HistPsyc Headlines Pages, which provide a history of important events related to psychology from 1650 to 1959, plus links to related sites. He has also developed several on-line study aids for students interested in the history of psychology that are available from his home page at www.unb.ca/web/units/psych/likely/psyc4053.htm.

Today in the History of Psychology

www.cwu.edu/~warren/today.html

Warren Street, at Central Washington University, offers a sample of events in the history of psychology for every day of the year. Pick a date, any date (as they say), from the History of Psychology Calendar and see what happened on that date. APA sponsors this site, which is based on Street's book, *A Chronology of Noteworthy Events in American Psychology*.

Scientific Methods in Psychology



2

★ MODULE 2.1

Science and the Evaluation of Evidence

Steps for Gathering and Evaluating Evidence

Replicability

Criteria for Evaluating Scientific Theories

The Principle of Parsimony and Degrees of Open-Mindedness

An Example of Applying Parsimony: Clever Hans, the Amazing Horse / Another Example of Applying Parsimony: Extrasensory Perception / Anecdotes / Experiments

★ MODULE 2.2

Methods of Investigation in Psychology

General Principles of Conducting Psychological Research

Definitions of Psychological Terms / Samples of the Population / Experimenter Bias and Blind Studies

Observational Research Designs

Naturalistic Observations / Case Histories / Surveys / Correlational Studies / The Correlation Coefficient / Correlation and Causation

Experiments

Independent Variables and Dependent Variables / Experimental Group, Control Group, and Random Assignment / STUDIES OF THE EFFECTS OF TELEVISED VIOLENCE ON AGGRESSIVE BEHAVIOR: WHAT'S THE EVIDENCE?

How Experiments Can Go Wrong: Demand Characteristics

Ethical Considerations in Experimentation

Ethical Concerns in Experiment on Humans / Ethical Concerns in Experiments on Animals

★ MODULE 2.3

Measuring and Analyzing Results

Descriptive Statistics

Measurements of the Central Score: Mean, Median, and Mode / Measures of Variation

Evaluating Results: Inferential Statistics

Every year spectacular claims are published about human behavior and related matters, even if little or no evidence exists to support the claims. Some examples:

- *Alien abductions.* Some people claim that they have been abducted by alien invaders, examined aboard a spacecraft, and then returned to Earth. To the question of why other people have not seen or photographed these aliens or their spacecraft, some of the “abductees” reply that the invaders and their craft are invisible, and so are the abductees during the period of their abduction.
- *Age regression.* Some people claim that, under hypnosis and using similar techniques, a person can recall in great detail what it was like to be a young child, a baby, an embryo, or even a sperm cell (Sadger, 1941). (One man said that he and his fellow sperm cells had resented their father because they knew he did not want them to fertilize the egg!)
- *Buried treasures.* According to one group of believers, a vault buried beneath Bruton Parish Church in Williamsburg, Virginia, contains “the missing crown jewels of England,” Francis Bacon’s birth certificate, the original manuscripts of Shakespeare’s plays, and some documents that can establish world peace (Sheaffer, 1992). Leaders of this

church had to get a legal restraining order to make these people stop digging under their church. One point that the believers cite as “evidence” for their claims: Look at the flag of the state of Virginia. It shows the Greek goddess Athena pointing a spear. And where is the spear pointed? *At the ground! Where the vault is buried!*

Most of us quickly dismiss such preposterous claims. Even if we do not dismiss them immediately, we are likely to ask the correct skeptical questions: What is the evidence? How good is that evidence? Can we find another, more reasonable explanation?

It is more difficult to discipline ourselves to ask these same questions about more plausible claims. When someone announces a “new scientific finding” that agrees with what we already believe or want to believe, we may find it very tempting to accept this new finding without subjecting it to careful scrutiny. It is a good idea to habitually check the evidence carefully for every claim, so far as possible; occasionally, the evidence will force us to change our opinions. This chapter concerns the kinds of evidence that psychologists use and the methods that they use to evaluate theoretical claims.

Science and the Evaluation of Evidence

How do scientists evaluate theories?

Why are most scientists so skeptical of new theories and claims that seem to contradict our current understanding?

You will sometimes hear nonscientists say that something has been “scientifically proved.” Scientists themselves seldom use the word *proved*, except when they are talking about a mathematical proof. As scientists collect more and better evidence, they may become confident about a given conclusion, but the word *prove* sounds a little too final.

One distinguishing characteristic of science is that scientists generally agree on how to evaluate competing theories. Even when they disagree about which theory is best, they can usually agree on what kinds of evidence they will accept. Most psychologists are quick to concede that our knowledge of psychology is less complete and less systematic than our knowledge of physics, chemistry, and biology. But like physicists, chemists, and biologists, psychologists do generally agree on what constitutes good evidence and what does not. They try to rely on the best available evidence and draw no conclusion at all if the evidence is weak.

SOMETHING TO THINK ABOUT

If people interested in ethics agreed with one another about how to evaluate theories, could they make progress comparable to that of scientists? Could theologians? ★

Steps for Gathering and Evaluating Evidence

Above all, scientists want to know the evidence behind a given claim. In psychology, as in other fields, students should learn to question assertions, to ask for the evidence behind a given claim.

The word *science* derives from a Latin word meaning *knowledge*, and there are many routes to scientific knowledge. The simplest is just careful observation, and much of science consists of recording observations. We also gain scientific knowledge by testing **hypotheses**, which are *testable predictions of what will happen under certain conditions*. Research designed to test hypotheses goes through the series of steps described in the following four paragraphs (see also Figure 2.1). Articles in scientific publications generally follow this sequence, too. In each of the following chapters of this book, you will find an example of a psychological study, described in a section entitled “What’s the Evidence?”

Hypothesis A hypothesis can be based on a larger theory. For example, “if our understanding of social influence is correct, then children who watch a great deal of violence will themselves become more violent.” In other cases, the hypothesis is the product of preliminary observations. For example, a psychologist might notice that several children who have outbursts of violent behavior have a habit of watching violent television programs and therefore suggest the hypothesis that watching violent programs leads to violent behavior.

Method Researchers have many methods for testing hypotheses. To test the effects of violent television shows, one possibility would be to measure how much time various children watch violence on television and relate that amount of time to a measure of their violent behavior. However, even if the correlation appeared strong, the results would not demonstrate cause and effect. (Maybe watching violence provokes violence, but it is also possible that children who are predisposed to violence like to watch it on TV.) Another approach would be to ask one group of children to

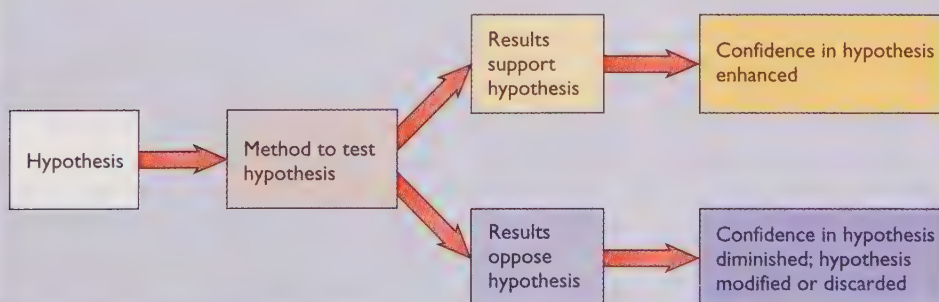


FIGURE 2.1 A hypothesis leads to predictions. An experimental method tests those predictions; a confirmation of a prediction supports the hypothesis; a disconfirmation indicates a need to revise or discard the hypothesis. Conclusions remain tentative, especially after only one experiment. Most scientists avoid saying that their results “prove” a conclusion.

watch one set of programs and another group to watch a different set of programs, and then record behavioral differences between the groups.

Results Fundamental to any research is good measurement, and a phenomenon such as “violent behavior” can be especially tricky to measure. (How do we decide what is *real* violence and what is just playfulness? Do threats count? verbal abuse?) It is important for an investigator to adopt clear rules for making measurements and then use these rules consistently. Depending on the nature of the study, the investigator calculates appropriate statistics for expressing the results and evaluating whether these probably indicate a meaningful trend or just a random fluctuation.

Interpretation The final task is to determine the significance of the results. If the results clearly contradict the hypothesis, researchers should either abandon or modify the original hypothesis. (Maybe it applies only to certain kinds of people or only under certain circumstances.) If the results *match* the prediction, investigators may gain confidence in their hypothesis but they should not necessarily accept it. Even though the results fit the hypothesis, they might also fit other hypotheses or explanations as well. Because almost any study has limitations, the ultimate conclusion can come only from a pattern of results derived from many studies.

Replicability

Years ago, investigators trained rats to respond in a certain way, then ground up the rats' brains, injected an extract of the trained rats' brains into other rats, and reported that the new rats remembered what the old rats had learned (Babich, Jacobson, Bubash, & Jacobson, 1965). Many other experimenters tried to replicate this surprising result. A few reported results somewhat similar to those of the first experimenters, but most investigators could not (Gaito, 1976; L. T. Smith, 1975). That is, these results were not replicable. **Replicable results** are *those that anyone can obtain, at least approximately, by following the same procedures.*

What should we do when results are not replicable? First, we determine whether the different investigators really used the same procedures. Sometimes, what appears to be a minor change in procedure yields a major difference in results. Did the researchers use different kinds of rats, or different methods of training, or different ways of extracting brain chemicals? In the research on transfer of training by brain extracts, psychologists found no consistent relationship between the results and any aspect of the procedure. In fact, most laboratories found no evidence that the brain extracts had any influence on other rats under any conditions. So, then what? The rule is that, if researchers cannot find conditions under which they dependably get a particular result, then they do not accept that result. This rule may

seem unduly harsh, but it is our best defense against error. If scientists cannot replicate a result, then the studies claiming to have that result may have hidden flaws.

Sometimes, however, an effect is small but real. For example, many studies have compared unprovoked aggressive behavior in men and women. Some studies have found that men are substantially more aggressive; some have found men to be slightly more aggressive; and a few have found no difference or a difference in the opposite direction. Despite this variation, we do not regard the sex difference in aggressive behavior to be unreplicable. Rather, our interpretation is that the sex difference is small and that it competes with many other influences on aggressive behavior. If we combine the results of many studies, we can get an approximate measure of the size of the effect. A **meta-analysis combines the results of many studies and analyzes them as though they were all one very large study.** For example, one meta-analysis found that men are indeed more likely than women to engage in unprovoked violence (Bettencourt & Miller, 1996). In most cases, a meta-analysis will also determine which variations in procedure are associated with the largest effects.

Criteria for Evaluating Scientific Theories

Up to now, I have alluded to research in psychology without using much detail. We shall consider the details of research methods later. Here, let's look at the big picture: After investigators collect mounds of evidence, what do they do with it?

One goal of scientific research is to establish **theories, comprehensive explanations of observable events.** A good theory predicts many observations in terms of a few assumptions and thus reduces the amount of information we must keep available for reference. For example, according to the *law of effect* (to be discussed in Chapter 6), if a human or any other animal makes a response that is consistently followed by a reinforcer (such as food to a hungry person or water to a thirsty one), then the future probability of that response will increase. This law summarizes results achieved for many species, many responses, and many reinforcers.

When we are confronted with several competing theories, we must evaluate them to decide which is the most acceptable (Figure 2.2). To illustrate, let's consider some unsatisfactory theories. First, consider what is wrong with this theory: “Karl is stingy, so he must have had a fixation in the anal period of psychosocial development.” The theory fits the data (Karl is stingy), but we already knew those data before the theorist said anything. No one had any other reason to believe that Karl had an anal fixation, except that he became stingy. So the theory does not really make any predictions. One important criterion for any theory is that it *should predict new observations.* If it accurately predicts ob-